

Contributions

1. **LongAudioQA**: first large-scale long-form audio meeting QA benchmark with 16,456 Q&A pairs across three corpora.
2. **GRGA**: training-free agent that models heterogeneous audio as a multi-dimensional graph and plans iteratively.
3. **Evaluation**: automatic + human studies show consistent gains over Speech LLM and RAG baselines.

Limitations & Future Work

- Noisy ASR/diarization can still corrupt graph evidence.
- Iterative planning is slower than single-turn RAG.
- Future: streaming QA and broader long-form audio domains.

Long-form Audio QA

Acoustic Missing + Context Fragmentation



LongAudioQA Dataset

3 corpora, 5 types, 16K+ Q&A



GRGA Model

Multi-dim Graph + POMDP Agent



SOTA Performance

+10% on AMI, +19.3% Citation P

Code & Data:

github.com/tangquanwei/GRGA